

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

192421406

I S-Jagadeesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S-Jagadeesh

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

④ IPv4 Subnetting

→ IP Address block = 172.16.10.0/24

→ Subnet mask = 127 (255.255.255.224)

Step 1 - number of subnets

→ Borrowed bits = $27 - 24 = 3$

→ Total subnets = $2^3 = 8$

Step 2 - usable hosts per subnet

→ Host bits = $32 - 27 = 5$

→ Total hosts = $2^5 = 32$

→ Usable hosts = $32 - 2 = 30$

Step 3 - Find the subnet of 172.16.10.178

- 172.16.10.0 — 31
- 172.16.10.32 — 63
- 172.16.10.64 — 95
- 172.16.10.96 — 127
- 172.16.10.128 — 159
- 172.16.10.160 — 191
- 172.16.10.192 — 223
- 172.16.10.224 — 255

172.16.10.78 belongs to subnet 172.16.10.64/27

Network address: 172.16.10.64

Broadcast address: 172.16.10.95

Usable Host range: 172.16.10.65 - 172.16.10.94

Step 4 - Check IP address 172.16.10.95

→ 172.16.10.95 is the broadcast address of the same subnet

→ It belongs to the same subnet but cannot be assigned to a host.

⑤ IP v6 Addressing

Given IPv6 Address

2001:0db8:0000:0000:0000:4400:0042:8329

Step 1 - compressed form

→ remove leading zero

→ replace consecutive zero group with ::

compressed Address:

2001:db8::4400:42:8329

Step 2 - Expanded form

2001:0db8:0000:0000:0000:4400:0042:8329

Step 3 - Network prefix (/64)

→ First 64 bits = Network prefix

Network prefix:

2001:0db8:0000:0000::/64

Step 4 - Total Host Address

→ Host bits = $128 - 64 = 64$

→ Total possible host addresses = 2^{64}

→ = 18,446,744,073,709,551,616 host addresses.